8b. Defining Type Stability

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INTRODUCTION

One of the primary reasons Julia achieves high performance is its ability to generate specialized machine code for functions. Thus, instead of relying on generic instructions, Julia can tailor code to the concrete operations and types appearing in a function. This capability relies heavily on **type inference**, the process by which the compiler deduces the types of variables and intermediate results before execution. When the compiler can successfully predict the concrete types of all operations within a function body, the resulting code is described as **type stable**.

Type stability isn't just a mere implementation detail, but rather a practical requirement for writing efficient Julia programs. Without it, the compiler must fall back to generic code, leading to significant performance penalties.

This section formally defines type stability and reviews the tools employed for its verification. In the next section, we'll start to examine how type stability applies in specific scenarios.

AN INTUITION

When a function is called, Julia follows a well-defined sequence of steps to determine how that call should be executed. We already **described this process** and now briefly review it.

Consider a function `foo(x) = x + 2` and the execution of `foo(a)` for some variable `a`. We assume `a` has a specific value assigned and therefore a concrete type, although we omit explicitly specifying a value for `a`. This lets us highlight that the process unfolded depends on types, rather than values.

When evaluating `foo(a)`, Julia first determines the concrete type of `a`, which we'll denote as `T`. It then checks whether a compiled method instance of `foo` specialized for an argument of type `T` already exists. If such an instance exists, then `foo(a)` is executed immediately. Otherwise, Julia proceeds to compile one. This compilation step leverages type inference, wherein the compiler attempts to deduce concrete types for all terms within the function body. The resulting machine code is then stored, making it readily available for subsequent calls of `foo(b)` with `b` having type `T`.

TYPE STABILITY AND PERFORMANCE

The key to generating fast code lies in the information available to the compiler during the compilation stage. This information is primarily gathered through type inference, where the compiler identifies the specific type of each variable and expression involved. When the compiler can **accurately predict a single concrete type for the function's output**, the function call is said to be **type stable**.

While this constitutes the **formal definition of type stability**, a more stringent definition is usually applied in practice: the compiler must be able to **infer concrete types for *each* expression within the function**, rather than only for the final output. This definition aligns with the output provided by `@code_warntype`, which is the built-in macro to detect type instabilities. The use of this macro we'll be presented in the next subsection.

When type stability holds, the compiler can specialize the computational approach for each operation, resulting in fast execution. Essentially, type stability dictates that there's sufficient information to determine a straight execution path, thus avoiding unnecessary type checks and dispatches at runtime.

In contrast, type-unstable functions generate generic code, thus accommodating each possible combination of concrete types. This results in additional overhead during runtime, where Julia is forced to dynamically gather type information and perform extra calculations based on it. The consequence is a pronounced deterioration in performance.

> **Type Stability Characterizes Function *Calls***
>
> It's common to describe a function as "type stable". Strictly speaking, however, type stability isn't a property of the function. Rather, it's a property of how the function behaves when called with arguments of particular concrete types. The distinction is crucial in practice, since a function may exhibit type stability for certain input types, but not for others.

AN EXAMPLE

To identify type stability in practice, let's consider the following example.

TYPE STABLE

```julia
x      = [1, 2, 3]          # `x` has type `Vector{Int64}`

foo(x) = sum(x[1:2])        # type stable with this `x`
```

```julia
julia> @btime foo($x)
  23.485 ns (2 allocations: 80 bytes)
```

TYPE UNSTABLE

```julia
x      = [1, 2, "hello"]    # `x` has type `Vector{Any}`

foo(x) = sum(x[1:2])        # type UNSTABLE with this `x`
```

```julia
julia> @btime foo($x)
  47.666 ns (2 allocations: 80 bytes)
```

The two operations may look identical at first glance, since both ultimately evaluate `1 + 2`. However, the way Julia compiles and executes each version differs. In fact, the first approach is faster since the function is type stable.

In that first version, the expression `x[1] + x[2]` can be inferred to have type `Int64`. Because the compiler can determine that both `x[1]` and `x[2]` are `Int64`, it's able to generate efficient, specialized machine code for this concrete type. Importantly, this optimization isn't limited to the specific `x` used: any call of the form `sum(y)` where `y` is a `Vector{Int64}` benefits from the same specialization.

The second version, however, introduces type instability. Here, `x` has type `Vector{Any}`, which prevents the compiler from deducing a single concrete type for the expression `x[1] + x[2]`. Since the elements of a `Vector{Any}` may hold values of any subtype of `Any`, the compiler can't predict whether `x[1]` or `x[2]` will be an `Int64`, a `Float64`, a `Float32`, or other type. This results in slow code, and the penalty applies to every call of `sum(y)` where `y` is a `Vector{Any}`.

> **Remark**
>
> Julia's developers are continually refining the compiler, addressing and mitigating the effects of certain type instabilities. As a result, **many operations that were once type unstable are now type stable**. This means that type stability should be considered a dynamic property of the language, subject to change as the compiler evolves.

CHECKING FOR TYPE STABILITY

There are several mechanisms to determine whether a function call is type stable. One of them is based on the `@code_warntype` macro, which reports all the types inferred during a function call. To illustrate its use, consider a function that defines `y` as a transformation of `x`. Then, `y` is used to perform some operation.

TYPE UNSTABLE

```
function foo(x)
    y = (x < 0) ?  0  :  x

    return [y * i for i in 1:100]
end
```

```
julia> @code_warntype foo(1.0)
```

TYPE STABLE

```julia
function foo(x)
    y = (x < 0) ? 0 : x

    return [y * i for i in 1:100]
end
```

```
julia> @code_warntype foo(1)
```

The output of `@code_warntype` can be difficult to interpret. Nonetheless, the inclusion of colors facilitates its understanding:

- If all lines are **blue**, the function is **type stable**. This means that Julia can identify a unique concrete type for each expression.
- If at least one line is **red**, the function is **type unstable**. It reflects that one or more expressions could potentially adopt multiple types.
- **Yellow** lines indicate type instabilities that the compiler can handle effectively, in the sense that they have a reduced impact on performance. As a rule of thumb, **you can safely ignore them**.

> ⚠️ **Warning!**
> Throughout the book, we'll refer to **type instabilities** as those indicated by a red warning exclusively. Yellow warnings will be mostly ignored.

In the provided example, the compiler attempts to infer concrete types. This is done by identifying two pieces of information given `x`'s concrete type:

i) the type of `y`,

ii) the type of `y * i`, with `i` identified with type `Int64`. Implicitly, this also defines the type of `[y * i for i in 1:100]`.

The example clearly demonstrates that **the same function can be type stable or unstable depending on the types of its inputs**: `foo` is type stable when `x` has type `Int64`, but type unstable when `x` is `Float64`.

Specifically, in the scenario where `x = 1`, the compiler infers for *i)* that `y` can be equal to either `0` or `x`. Since both `0` and `1` are `Int64`, the compiler identifies a unique type for `y`, given by `Int64`. Regarding *ii)*, `y * i` also yields an `Int64`, as both `i` and `y` have type `Int64`. This determines that `[y * i for i in 1:100]` has type `Vector{Int64}`. Consequently, `foo(1)` is type stable, enabling Julia to invoke a tailored method for integers.

As for `x = 1.0`, the information for *i)* is that `y` could be either `0` or `1.0`. As a result, the compiler can't infer a unique type for `y`, which could be either `Int64` or `Float64`. The `@code_warntype` macro reflects this, identifying `y` as having type `Union{Float64, Int64}`. This ambiguity affects *ii)*,

forcing the compiler to consider approaches that handle both `Float64` and `Int64`, and hence preventing specialization. Overall, `foo(1.0)` is type unstable, which has a detrimental impact on performance.

> **Function Call Leverage Information on Types, Not Values**
>
> The conclusions regarding type stability wouldn't have changed if we had respectively considered `foo(-2)` or `foo(-2.0)` in each tab. Type stability depends on whether `x` has type `Int64` or `Float64`, regardless of its actual value. In other words, since the compiler analyzes type information exclusively, the actual numeric value plays no role in determining type stability.

YELLOW WARNINGS MAY TURN RED

Not all type instabilities carry the same performance cost, and Julia indicates their severity through yellow and red warnings. A yellow warning typically signals a mild issue: the compiler has detected a type instability, but it occurs in a small isolated computation that Julia can still optimize reasonably well. However, repeated execution of these operations may escalate into more serious performance issues, triggering a red warning. The following example demonstrates a scenario like this.

YELLOW WARNING

```julia
function foo(x)
    y = (x < 0) ? 0 : x

    y * 2
end
```

```
julia> @code_warntype foo(1.0)
```

RED WARNING

```julia
function foo(x)
    y = (x < 0) ? 0 : x

    [y * i for i in 1:100]
end
```

```
julia> @code_warntype foo(1.0)
```

YELLOW WARNING

```
function foo(x)
    y = (x < 0) ?  0  :  x

    for i in 1:100
      y = y + i
    end

    return y
end
```

```
julia> @code_warntype foo(1.0)
```

In the first case, the yellow warning appears because the expression `y * 2` may produce either a `Float64` or an `Int64`. Since this computation happens only once and involves types the compiler can handle efficiently, the impact on performance is minimal. In contrast, the second tab performs repeated evaluations of `y * i` without a concrete type for `y`, resulting in a red warning.

Note, though, that a yellow warning doesn't necessarily escalate into a red warning when incorporated into a for-loop. The third tab demonstrates a situation where the compiler can still manage the instability effectively, even under repeated execution. This highlights that type instabilities vary in severity, and not all of them pose a meaningful threat to performance.

For-Loops and Yellow Warnings

When running for-loops, Julia will always emit a yellow warning, even if the operation is type stable. The warning can safely be disregarded, as it simply reflects the inherent behavior of iterators: they return either the next element to iterate over, or `nothing` when the sequence is exhausted (a value with type `Nothing`).

YELLOW WARNING IN FOR-LOOPS

```
function foo()
    for i in 1:100
        i
    end
end
```

```
julia> @code_warntype foo()
```